

VoIP PBX Deployment using Asterisk and FreeBSD

Portfolio Project Documentation

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2026

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1. Introduction

Voice over Internet Protocol (VoIP) is a technology that enables voice and multimedia communication over IP networks instead of traditional telephone systems. Modern organizations widely use VoIP solutions due to their cost-effectiveness, scalability, and flexibility in supporting communication services.

This project focuses on the deployment and configuration of a VoIP Private Branch Exchange (PBX) system using Asterisk on FreeBSD. The implementation includes the installation and configuration of the Asterisk server, creation of SIP endpoints, call routing through dial plans, voicemail services, and call detail record (CDR) management. In addition, voice and video communication were tested using softphone applications to evaluate the functionality and reliability of the system.

The project also provides practical exposure to networking concepts such as SIP signaling, RTP media transmission, endpoint registration, and VoIP troubleshooting. Various testing and verification activities were performed to ensure successful communication between clients and proper operation of PBX services.

Overall, this project demonstrates the design, implementation, and validation of a functional VoIP environment while developing practical skills in Linux-based server administration, network communication, and enterprise telephony technologies.

2. Objectives

1. To have a practical experience related to Asterisk PBX.
2. In order to configure PJSIP endpoints and to register SIP clients
3. To verify voice and video calling between MicroSIP and Linphone.
4. To examine SIP and RTP traffic on CLI logs and Wireshark
5. To provide voicemail, call forwarding, call detail record and conferencing
6. To work out all features into a single final PBX.

3. System Setup

3.1 Hardware and Virtualisation

- **Host:** Windows 11 (Laptop)
- **Virtualization:** VirtualBox
- **VM Specs:** 2 vCPUs, 2 GB RAM, 16 GB disk
- **Network Mode:** Bridged Adapter

3.2 Software Components

- **Operating System:** FreeBSD (v14.3)
- **PBX:** Asterisk (v20)
- **Softphones:** MicroSIP (windows), Linphone (Windows)

3.3 Network Configuration

- **SIP Signaling:** UDP 5060
- **RTP Ports:** 10000 – 20000
- **IP Address (VM):** 192.168.43.4

4. Practical Implementation Weekly

4.1 Week 1 – Asterisk Installation

Installed Asterisk on FreeBSD using package manager. Verified CLI access with `asterisk -rvvv`.

```
192.168.43.4 - PuTTY
top -m io

-- Benedict Reuschling <bcr@FreeBSD.org>
dineth@freebsd:~ $
dineth@freebsd:~ $
dineth@freebsd:~ $ service asterisk start
asterisk already running? (pid=837).
dineth@freebsd:~ $ asterisk -r
Unable to connect to remote asterisk (does /var/run/asterisk/asterisk.ctl exist?)
dineth@freebsd:~ $ su
Password:
root@freebsd:/home/dineth # sudo asterisk -rvvv
Asterisk 20.15.2, Copyright (C) 1999 - 2025, Sangoma Technologies Corporation and others.
Created by Mark Spencer <markster@digium.com>
Asterisk comes with ABSOLUTELY NO WARRANTY; type 'core show warranty' for details.
This is free software, with components licensed under the GNU General Public License version 2 and other licenses; you are welcome to redistribute it under certain conditions. Type 'core show license' for details.
=====
Connected to Asterisk 20.15.2 currently running on freebsd (pid = 837)
freebsd*CLI> 
```

4.2 Week 2 PJSIP Configuration and show Endpoints

```
[6001](endpoint-basic)
```

```
callerid = "Dineth" <6001>
```

```
auth = 6001
```

[REDACTED]

[6001](aor-single-reg)

[REDACTED]

username = 6002

password = voip123

[6002](aor-single-reg)

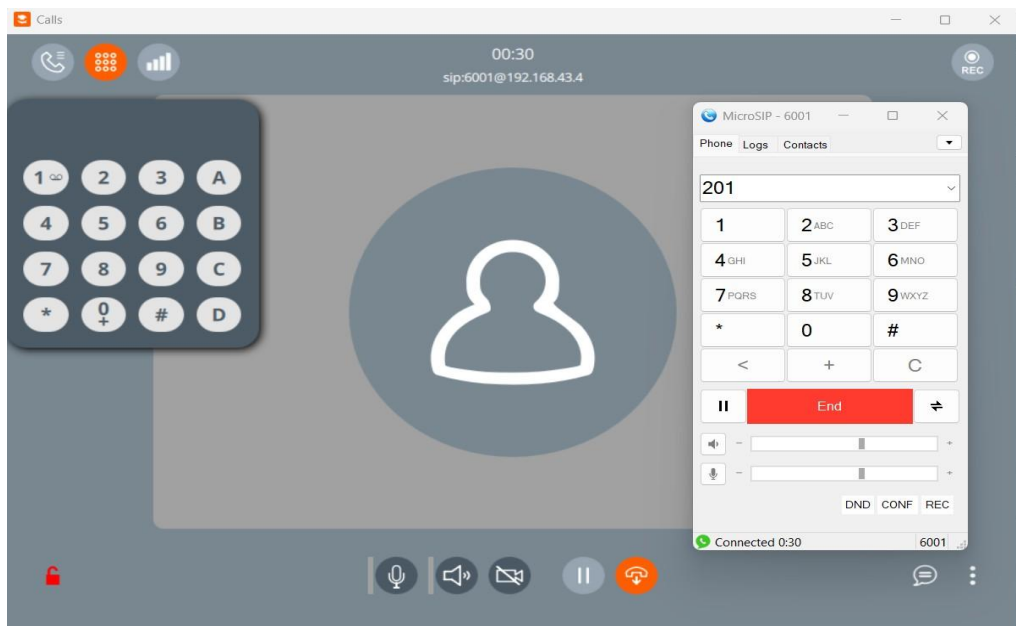
```
Endpoint: 6001/200                                Not in use    0
of inf
  InAuth: 6001/6001
    Aor: 6001                                     5
    Contact: 6001/sip:6001@192.168.43.124:57561;ob 8d928147e2 NonQual
      nan
  Transport: transport-udp      udp      0      0 0.0.0.0:5060

Endpoint: 6002/201                                Not in use    0
of inf
  InAuth: 6002/6002
    Aor: 6002                                     5
    Contact: 6002/sip:6002@192.168.43.124;transport=udp c4e32c4620 NonQual
      nan
  Transport: transport-udp      udp      0      0 0.0.0.0:5060

Objects found: 2
```

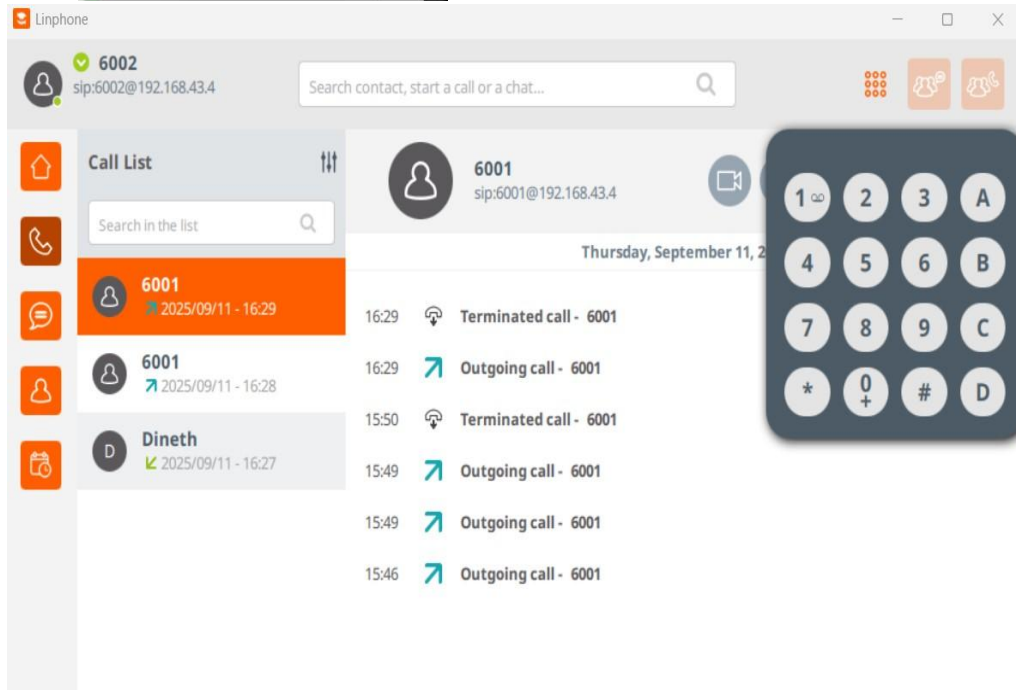
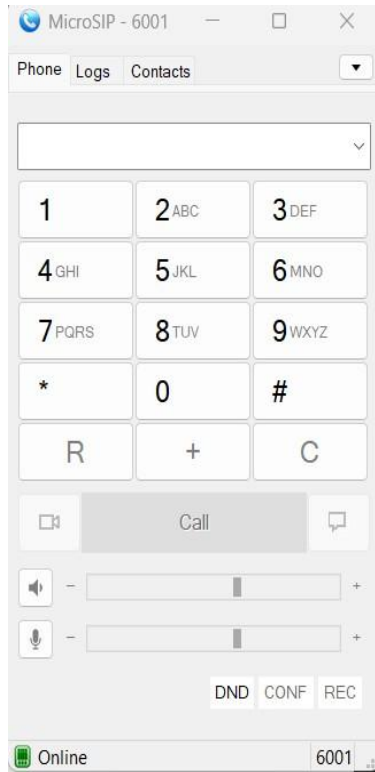
4.3 Week 3 -Dialplan and Basic Call Routing.

```
[fxsgroup]
```



4.4 Week 4 – Softphone Registration

Configured 6001 (MicroSIP) and 6002 (Linnphone). Verified registration using:
pjsip show endpoints



4.5 SIP Video Negotiation and Codecs.

(pjsip.conf)

```
[global]
type=global

[transport-udp]
type=transport
protocol=udp
bind=0.0.0.0

;-----
; - Endpoint Template -
;-----

[endpoint-basic](!)
type = endpoint
transport = transport-udp
context=fxsgroup
disallow=all
allow=ulaw
allow=slin
allow = h263
allow = h264
direct_media = no

[auth-userpass](!)
type = auth
```

```
auth_type = userpass
password = voip123
[aor-single-reg](1)
type = aor
max_contacts = 2
```

```
;-----x200
[6001](endpoint-basic)
callerid = "Dineth" <200>
auth = 6001
aors = 6001
```

```
[6001](auth-userpass)
username = 6001
```

```
[6001](aor-single-reg)
max_contacts = 5
;-----x201
```

```
[6002](endpoint-basic)
callerid = "Friend" <201>
auth = 6002
aors = 6002
```

```
[6002](auth-userpass)
username = 6002
```

```
[6002](aor-single-reg)
max_contacts = 5
```

Dialplan configuration (**Extensions.conf**)

```
[global]

[fxsgroup]
exten => 200,1,NoOp()
    same=>n,Dial(PJSIP/6001)
    same=>n,Congestion()
    same=>n,Hangup()

exten => 300,1,NoOp()
    same=>n,Answer()
    same=>n,Wait(2)
    same=>n,Playback(demo-congrats)
    same=>n,Wait(2)
    same=>n,Congestion()
    same=>n,Hangup()

exten => 400,1,Dial(PJSIP/6001)
exten => 400,1,Dial(PJSIP/6002)

exten => 100,1,Answer()
    same => n,Playback(hello-world)
    same => n,Hangup()

exten => 500,1,Answer()
    same => n,Playback(demo-congrats)
    same => n,Hangup()
```

4.6 Voicemail setup and retrieval

Configure Voicemail

```
[default]

6001 => 1234, MicroSIP User, microsip@example.com
6002 => 1234, Linphone User, linphone@example.com
```

Dialing 5000 allowed retrieval of stored messages after entering mailbox credentials.

```
GNU nano 8.4 /usr/local/etc/asterisk/voicemail.conf

[default]
; Note: The rest of the system must reference mailboxes defined here as mailbox@default.

1234 => 4242,Example Mailbox,root@localhost
6001 => 1234, MicroSIP User, microsip@example.com
6002 => 1234, Linphone User, linphone@example.com
; 4200 => 9855,Mark Spencer,markster@linux-support.net,mypager@digium.com,attach=no|servermai
; 4300 => 3456,Ben Rigas,ben@american-computer.net
; 4310 => -5432,Sales,sales@marko.net
; 4069 => 6522,Matt Brooks,matt@marko.net,,|tz=central|attach=yes|saycid=yes|dialout=fromvm|ca
; 4073 => 1099,Bianca Paige,bianca@biancapaige.com,,delete=1|emailsubject=You have a new voice
; 4110 => 3443,Rob Flynn,rfflynn@blueridge.net
; 4235 => 1234,Jim Holmes,jim@astricon.ips,,Tz=european

; Aliases allow alternate references to mailboxes. See the "aliasescontext"
; parameter in the "general" section.
;
[myaliases]
1234@devices => 1234@default
6200@devices => 4200@default

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Configured voicemail.conf with mailboxes for 6001 and 6002. Dialed 5000 from Linphone to retrieve voicemail. Messages were successfully recorded and played back

```

Connected to Asterisk 20.15.2 currently running on freebsd (pid = 901)
freebsd*CLI>
freebsd*CLI> core set verbose 5
Console verbose was OFF and is now 5.
-- Executing [6001@fxsgroup:1] Dial("PJSIP/6002-0000000a", "PJSIP/6001,20") in new stack
-- Called PJSIP/6001
-- PJSIP/6001-0000000b is ringing
== Everyone is busy/congested at this time (1:1/0/0)
-- Executing [6001@fxsgroup:2] VoiceMail("PJSIP/6002-0000000a", "6001@default,u") in new stack
> 0x3716ce487000 -- Strict RTP learning after remote address set to: 192.168.43.124:7078
-- <PJSIP/6002-0000000a> Playing 'vm-theperson.gsm' (language 'en')
> 0x3716ce487000 -- Strict RTP switching to RTP target address 192.168.43.124:7078 as source
-- <PJSIP/6002-0000000a> Playing 'digits/6.gsm' (language 'en')
-- <PJSIP/6002-0000000a> Playing 'digits/0.gsm' (language 'en')
-- <PJSIP/6002-0000000a> Playing 'digits/0.gsm' (language 'en')
> 0x3716ce487000 -- Strict RTP learning complete - Locking on source address 192.168.43.124:7078
-- <PJSIP/6002-0000000a> Playing 'digits/1.gsm' (language 'en')
-- <PJSIP/6002-0000000a> Playing 'vm-isunavail.gsm' (language 'en')
-- <PJSIP/6002-0000000a> Playing 'vm-intro.gsm' (language 'en')
-- <PJSIP/6002-0000000a> Playing 'beep.gsm' (language 'en')
-- Recording the message
-- Recording automatically stopped after a silence of 10 seconds
-- <PJSIP/6002-0000000a> Playing 'auth-thankyou.gsm' (language 'en')
-- Executing [6001@fxsgroup:3] Hangup("PJSIP/6002-0000000a", "") in new stack
== Spawn extension (fxsgroup, 6001, 3) exited non-zero on 'PJSIP/6002-0000000a'
-- Executing [5000@fxsgroup:1] VoiceMailMain("PJSIP/6002-0000000c", "") in new stack
> 0x3716ce487000 -- Strict RTP learning after remote address set to: 192.168.43.124:7078
-- <PJSIP/6002-0000000c> Playing 'vm-login.gsm' (language 'en')
> 0x3716ce487000 -- Strict RTP switching to RTP target address 192.168.43.124:7078 as source
> 0x3716ce487000 -- Strict RTP learning complete - Locking on source address 192.168.43.124:7078
-- <PJSIP/6002-0000000c> Playing 'vm-password.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-youhave.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'digits/2.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-INBOX.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-messages.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-onefor.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-INBOX.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-messages.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-opts.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-helpexit.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-onefor.gsm' (language 'en')
-- <PJSIP/6002-0000000c> Playing 'vm-INBOX.gsm' (language 'en')

```

4.7 Call Detail Records (CDR)

The following modules are expected `cdr_csv.so`

Enable CDR in `cdr.conf`

```
root@freebsd:/home/dineth # nano /usr/local/etc/asterisk/cdr.conf
GNU nano 8.4 /usr/local/etc/asterisk/cdr.conf

;
; Asterisk Call Detail Record engine configuration
;
; CDR is Call Detail Record, which provides logging services via a variety of
; pluggable backend modules. Detailed call information can be recorded to
; databases, files, etc. Useful for billing, fraud prevention, compliance with
; Sarbanes-Oxley aka The Enron Act, QOS evaluations, and more.
;
[general]
enable = yes

; Define whether or not to use CDR logging. Setting this to "no" will override
; any loading of backend CDR modules. Default is "yes".
enable=yes

; Define whether or not to use CDR logging on new channels by default.
; Setting this to "no" will disable CDR on channels unless it is explicitly
; enabled. Default is "yes".
channeldefaultenabled=yes
```

Mster.csv file (Call logs history)

```
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000004","PJSIP/6002-00000005","Dial","PJSIP/6002,20","2025-09-18 21:06:06","2025-09-18 21:06:14","2025-09-18 21:06:20",13,5,"ANSWERED","DOCUMENTATION","1758229566.8","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000006","PJSIP/6002-00000007","Dial","PJSIP/6002,20","2025-09-18 21:06:36","2025-09-18 21:06:56",20,0,"NO ANSWER","DOCUMENTATION","1758229596.11","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000006","","Hangup","""
"","2025-09-18 21:06:56","2025-09-18 21:06:56","2025-09-18 21:07:20",23,23,"ANSWERED","DOCUMENTATION","1758229596.11","""
"","2564286000","5000","fxsgroup","""MyName Here"" <2564286000>","Console/default","""
"","Hangup","""
"","2025-09-18 21:14:08","2025-09-18 21:14:08","2025-09-18 21:15:23",74,74,"ANSWERED","DOCUMENTATION","1758230048.15","""
"","2564286000","5000","fxsgroup","""MyName Here"" <2564286000>","Console/default","""
"","Hangup","""
"","2025-09-18 21:16:11","2025-09-18 21:16:11","2025-09-18 21:17:27",75,75,"ANSWERED","DOCUMENTATION","1758230171.17","""
"","200","5000","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000008","","VoiceMailMain","""
"","2025-09-18 21:20:01","2025-09-18 21:20:01","2025-09-18 21:20:34",32,32,"ANSWERED","DOCUMENTATION","1758230401.19","""
"","200","5000","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000009","","Hangup","""
"","2025-09-18 21:21:13","2025-09-18 21:21:13","2025-09-18 21:22:25",72,72,"ANSWERED","DOCUMENTATION","1758230473.21","""
"","201","6001","fxsgroup","""Friend"" <201>","PJSIP/6002-0000000a","PJSIP/6001-0000000b","Dial","PJSIP/6001,20","2025-09-18 21:31:16","2025-09-18 21:31:25",8,0,"BUSY","DOCUMENTATION","1758231076.23","""
"","201","6001","fxsgroup","""Friend"" <201>","PJSIP/6002-0000000a","","Hangup","""
"","2025-09-18 21:31:25","2025-09-18 21:31:25","2025-09-18 21:31:50",24,24,"ANSWERED","DOCUMENTATION","1758231076.23","""
"","201","5000","fxsgroup","""Friend"" <201>","PJSIP/6002-0000000c","","VoiceMailMain","""
"","2025-09-18 21:31:57","2025-09-18 21:31:57","2025-09-18 21:32:47",49,49,"ANSWERED","DOCUMENTATION","1758231117.27","""
"","201","5000","fxsgroup","""Friend"" <201>","PJSIP/6002-0000000d","","VoiceMailMain","""
"","2025-09-18 21:33:02","2025-09-18 21:33:02","2025-09-18 21:34:55",113,113,"ANSWERED","DOCUMENTATION","1758231182.29","""
"","201","5000","fxsgroup","""Friend"" <201>","PJSIP/6002-0000000e","","VoiceMailMain","""
"","2025-09-18 22:11:49","2025-09-18 22:11:49","2025-09-18 22:12:31",42,42,"ANSWERED","DOCUMENTATION","1758233508.31","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-0000000f","PJSIP/6002-00000010","Dial","PJSIP/6002,20","2025-09-18 22:17:45","2025-09-18 22:17:53",8,0,"NO ANSWER","DOCUMENTATION","1758233865.33","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-0000000f","","Hangup","""
"","2025-09-18 22:17:53","2025-09-18 22:17:53","2025-09-18 22:18:17",23,23,"ANSWERED","DOCUMENTATION","1758233865.33","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000000","PJSIP/6002-00000001","Dial","PJSIP/6002,20","2025-09-19 10:08:23","2025-09-19 10:08:27",3,0,"NO ANSWER","DOCUMENTATION","1758276503.0","""
"","200","6002","fxsgroup","""Dineth"" <200>","PJSIP/6001-00000002","PJSIP/6002-00000003","Dial","PJSIP/6002,20","2025-09-19 10:08:53","2025-09-19 10:08:57",2025-09-19 10:09:30",37,33,"ANSWERED","DOCUMENTATION","1758276533.3","""
root@frebsd:/home/dineth #
```

5. Result and Evidence

Results Evidence Provide	Task	Status
Softphone Registration	CLI Logs + Screenshots	Working
Voice Calls	Call logs (6001 <-> 6002)	Working
Video Calls	SDP negotiation captured	Working
Voicemail	Message retrieval tested	Working
Call Forwarding	Verified via Extensions.conf	Working
Call Records (CDR)	Entries in Master.csv	Working

6. Conclusion

By the end of the lab series, a fully functional PBX system was implemented with end-to-end voice and video support, voicemail, call detail records, call forwarding, and conferencing. This demonstrated the versatility of Asterisk as an open-source PBX solution and provided deep practical insights into SIP signaling, RTP streams, and VoIP system administration.

The experiments successfully bridged theoretical VoIP concepts with hands-on implementation, preparing students for real-world telecommunication environments.

7. References

1 Asterisks -

- Install FreeBSD and deploy asterisk on FreeBSD OS:
<https://youtu.be/t5DSr5efTZQ?si=ej3iDu2PHAxWz67t>

2 LinePhone Documentation

- Configuring audio/video SIP clients on Windows.
<https://wiki.linphone.org/xwiki/wiki/public/view/Main/>

8. Final Video Output

 Video Link here - <https://youtu.be/t3nbT1BzfAk?si=cA1YEla8MTIRe3pc>

-The END-